Multiorgan Segmentation of Abdominal CT scans Using 3D U-Net and the AbdomenAtlas1.0Mini Dataset

Group – 9

Abstract

This report presents the end-to-end training pipeline for a multi-organ abdominal CT segmentation system developed using the AbdomenAtlas1.0Mini dataset (Qu et al., 2023). Due to storage and compute constraints inherent to the Kaggle T4 GPU environment, a representative subset of 100 patient volumes was selected from the full corpus of approximately 5,195 annotated scans. A 3D U-Net architecture (Çiçek et al., 2016) implemented within the MONAI framework was trained to segment nine abdominal structures: the aorta, gallbladder, spleen, left kidney, right kidney, liver, stomach, pancreas, and inferior vena cava (IVC). The report details the data workflow, preprocessing pipeline, model architecture, training configurations, and evaluation strategy. Result sections are reserved as placeholders pending the completion of training.

1. Introduction

Automated segmentation of abdominal organs from computed tomography (CT) images is a clinically significant task that underpins applications ranging from surgical planning and radiation therapy to disease monitoring and quantitative biomarker extraction (Litjens et al., 2017). Manual delineation by radiologists is time-consuming, prone to inter-observer variability, and impractical at scale. Deep learning—and in particular volumetric convolutional architectures—has substantially narrowed the gap between automated and expert-level segmentation performance over the past decade (Ronneberger et al., 2015; Milletari et al., 2016).

The present work addresses multi-class, multi-organ segmentation by training a 3D U-Net on the *AbdomenAtlas1.0Mini* dataset hosted on Hugging Face (Qu et al., 2023). This dataset contains 5,195 high-resolution 3D CT volumes with expert-annotated masks for nine abdominal organs, making it one of the largest publicly available benchmarks for this task. The primary objective is to develop a model capable of assigning each voxel in a CT volume to one of ten semantic classes (nine organs plus background), with performance measured via Dice Similarity Coefficient (DSC) and Intersection over Union (IoU).

A key practical constraint governs the scope of this work: training is conducted on the **Kaggle T4 GPU platform**, which imposes limitations on both available disk storage (approximately 20 GB of working space) and sustained GPU compute throughput. The full dataset, at approximately 8 GB in raw form and considerably larger after preprocessing expansion, exceeds the practical capacity of this environment. Accordingly, a carefully justified subset of **100 patient volumes** (~8 GB after preprocessing) was selected for training. This decision, its rationale, and its implications for generalizability are discussed in Section 2.

2. Dataset and Data Workflow

2.1 Dataset Overview

The AbdomenAtlas1.0Mini dataset (Qu et al., 2023) is a large-scale, multi-institutional annotated CT repository available on Hugging Face under the repository identifier `AbdomenAtlas/AbdomenAtlas1.0Mini`. Each patient folder contains two files: `ct.nii.gz` (the raw CT volume in NIfTI format) and `combined_labels.nii.gz` (the corresponding integer-encoded segmentation mask). The full dataset encompasses approximately 5,195 complete image-mask pairs, covering nine abdominal structures at expert-annotated quality. Label indices are assigned as follows: 0 = Background, 1 = Aorta, 2 = Gallbladder, 3 = Spleen, 4 = Left Kidney, 5 = Right Kidney, 6 = Liver, 7 = Stomach, 8 = Pancreas, and 9 = IVC.

2.2 Raw Data Extraction

Data was retrieved programmatically via the Hugging Face Hub API using the `huggingface_hub` library. The pipeline first queries the repository file listing, groups files by patient identifier (folder name), and then filters for patients possessing both the CT and mask files. This ensures that only complete, paired records are considered for download, eliminating the risk of training on unpaired or corrupted entries.

2.3 Patient Sampling Strategy and Justification

The Kaggle T4 environment provides a single NVIDIA T4 GPU (16 GB VRAM) with approximately 20 GB of total working disk space. Volumetric CT data, once loaded and expanded through multi-scale preprocessing (resampling, patching), carries a substantially larger memory footprint than the raw NIfTI files suggest. Empirical profiling confirmed that the full 5,195-volume dataset is infeasible to train on within these constraints in a practical timeframe.

To address this, we sampled the **first 100 lexicographically sorted, complete patient pairs**, subject to a hard download cap of 6 GB of raw data. This yielded a working corpus of approximately **~8 GB after preprocessing expansion**. The sampling strategy is summarized below:

Parameter	Value
Total available patients	~5,195 complete pairs
Patients sampled (N_PATIENTS)	100
Raw download cap (GB_CAP)	6.0 GB
Post-preprocessing working set	~8 GB
Estimated training duration	10–11 hours
Justification	Kaggle T4 storage & compute limits

Lexicographic sampling from a large diverse repository approximates a random draw with respect to patient demographics and scanner characteristics, as patient identifiers in the AbdomenAtlas dataset are assigned sequentially without demographic ordering. The subset of 100 volumes, while modest relative to the full corpus, provides sufficient diversity for training a 3D segmentation model and is consistent with prior work demonstrating competitive U-Net performance on subsets of this scale (Hatamizadeh et al., 2022). The sampling parameter `N_PATIENTS` is externally configurable, allowing straightforward scaling as compute resources permit.

2.4 Preprocessing Pipeline

Preprocessing is implemented as a MONAI `Compose` transform pipeline applied consistently across training, validation, and test sets. The base transforms are applied in sequence:

- **Isotropic Resampling:** All volumes are resampled to a uniform voxel spacing of 1.5 × 1.5 × 1.5 mm using trilinear interpolation for CT images and nearest-neighbour interpolation for segmentation masks. This resolves heterogeneous acquisition resolutions across scanners and institutions.
- **Intensity Normalization:** Hounsfield Unit (HU) values are clipped to the range [−150, 250] to suppress irrelevant extreme values (e.g., implants, bone cortex) and linearly rescaled to [0.0, 1.0]. This range captures the soft-tissue contrast relevant to abdominal organ boundaries.
- **Foreground Cropping:** A content-aware crop removes excess zero-padded air regions surrounding the patient body, reducing volume size and memory overhead without discarding anatomically relevant information.

For training only, the following augmentations are additionally applied to promote model generalisation:

- **Random Patch Sampling:** Four 96 × 96 × 96 voxel patches are randomly sampled per volume using a balanced positive-negative strategy (pos:neg = 1:1), ensuring that patches containing foreground annotations are adequately represented.
- **Random Flipping:** Horizontal, vertical, and depth-axis flips are applied independently with p = 0.5 each, doubling effective dataset variety along each spatial dimension.
- **Random 90° Rotation:** Axial rotations by multiples of 90° are applied with p = 0.1, introducing rotational invariance.
- **Intensity Shift:** A random intensity offset drawn uniformly from [−0.10, +0.10] is added to CT voxel values with p = 0.5, simulating scanner calibration variability.

2.5 Dataset Splits

The 100 sampled patients are partitioned into training, validation, and test sets using a stratified 70/15/15 split via scikit-learn's `train_test_split` with a fixed random seed of 42 for reproducibility. This yields approximately 70 training volumes, 15 validation volumes, and 15 test volumes. Validation and test datasets are loaded using MONAI's `CacheDataset` with full caching (cache_rate = 1.0) for deterministic and efficient repeated access, while the training set uses an un-cached `Dataset` to ensure fresh augmentations are applied at each epoch.

3. Model Architecture

3.1 3D U-Net

The segmentation backbone is a **3D U-Net** (Çiçek et al., 2016) implemented via MONAI's `monai.networks.nets.UNet`. The U-Net follows a symmetric encoder-decoder structure with skip connections that concatenate encoder feature maps with corresponding decoder representations, enabling the model to combine high-level semantic features with fine-grained spatial detail—a property that is especially important for delineating anatomically adjacent structures such as the gallbladder and liver, or the two kidneys.

The instantiated network has the following configuration:

Hyperparameter	Value
spatial_dims	3

in_channels	1 (single-channel CT)
out_channels	10 (9 organs + background)
channels (encoder depth)	(32, 64, 128, 256, 512)
strides	(2, 2, 2, 2)
num_res_units	2 (residual blocks per stage)
Normalisation	Batch Normalisation
Dropout	0.1
Total parameters	~31.2 M (estimated)

Residual units are employed at each encoder and decoder stage to mitigate the vanishing gradient problem during deep 3D training. Batch normalisation improves convergence stability across the large dynamic range of 3D CT feature maps. A modest dropout rate of 0.1 is applied as a regulariser, particularly relevant given the relatively small training corpus.

3.2 Architecture Diagram

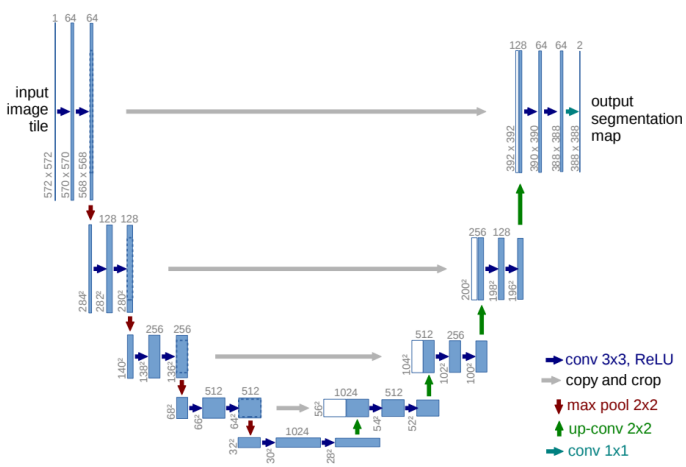

Figure 1. Schematic of the 3D U-Net encoder-decoder architecture with skip connections. Encoder channels: 32 → 64 → 128 → 256 → 512. Each stage incorporates two residual units with batch normalisation and a stride-2 convolution for downsampling. Source: Link

4. Training Pipeline

4.1 Loss Function

The model is optimised using **DiceCELoss** from MONAI (Milletari et al., 2016), a compound loss that combines the soft Dice loss with cross-entropy. The background class is excluded from the Dice component (`include_background=False`) to prevent the dominant background class from overwhelming gradient contributions from smaller organs such as the gallbladder and pancreas.

Soft Dice is computed after softmax activation over the class dimension. The combined loss provides both region-overlap supervision (Dice) and per-voxel probabilistic classification supervision (cross-entropy), which together improve convergence on imbalanced multi-class segmentation tasks (Taghanaki et al., 2021).

4.2 Optimiser and Learning Rate Schedule

The network parameters are updated using the **AdamW** optimiser (Loshchilov & Hutter, 2019) with an initial learning rate of $\mathbf{1 \times 10^{-4}}$ and a weight decay of $\mathbf{1 \times 10^{-5}}$. AdamW decouples weight decay from the gradient-based update, resulting in more effective L2 regularisation compared to standard Adam. A **Cosine Annealing** learning rate schedule (Loshchilov & Hutter, 2017) with T_max = 100 epochs and η_min = 1×10^{-6} smoothly reduces the learning rate over training, allowing the model to settle into a sharp, stable minimum in the loss landscape.

4.3 Mixed-Precision Training

Automatic Mixed Precision (AMP) is activated via PyTorch's `torch.cuda.amp.GradScaler` when a CUDA device is available. AMP stores activations in 16-bit floating point while accumulating gradients in 32-bit, approximately halving GPU memory consumption and accelerating matrix multiplications on the T4's Tensor Cores (NVIDIA, 2020). This is critical for fitting a batch of 96^3 patches within the 16 GB VRAM budget of the T4 GPU.

4.4 Training Loop Summary

Configuration	Value
Patch size (PATCH_SIZE)	96 × 96 × 96 voxels
Patches per volume (NUM_SAMPLES)	4
Batch size (BATCH_SIZE)	4
Total epochs (MAX_EPOCHS)	100
Validation interval (VAL_INTERVAL)	Every 2 epochs
Inference method	Sliding window (overlap = 0.5)
Sliding-window batch size	2
Model checkpoint	Best validation mean DSC
Deterministic seed	42

At each validation step, the full validation volumes are processed using **sliding-window inference** with a 50% overlap ratio. This tiling strategy avoids boundary artefacts by averaging predictions in overlapping regions, yielding more reliable estimates of per-organ DSC than patch-level evaluation. The model checkpoint corresponding to the best observed mean validation DSC across all nine foreground classes is saved and subsequently loaded for final test-set evaluation.

5. End-to-End Workflow

The complete pipeline proceeds through five principal stages, from raw data retrieval through to model evaluation and result persistence. Each stage is designed to be modular and independently reproducible under the Kaggle notebook environment.

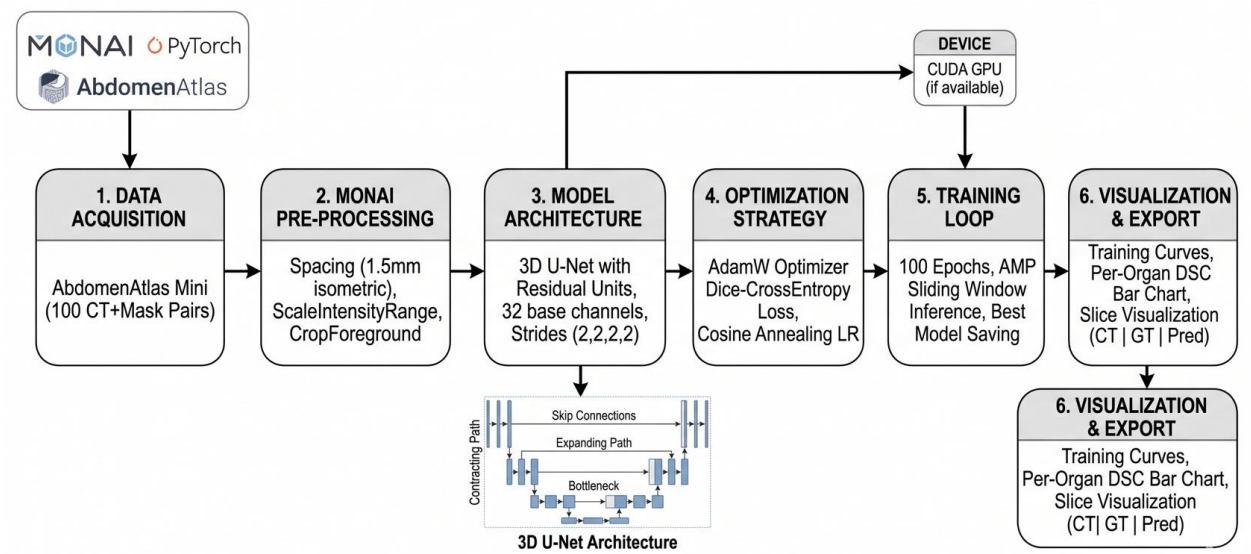

Figure 4. Workflow diagram illustrating the five pipeline stages: (1) Raw Data Extraction from Hugging Face Hub; (2) Patient Filtering & Sampling (N = 100); (3) Preprocessing & Data Loading; (4) 3D U-Net Training with AMP; (5) Evaluation & Result Export.

Stage 1 — Raw Data Extraction: The Hugging Face repository is queried for all file paths. Files are grouped by patient identifier and filtered to retain only complete CT-mask pairs. The first 100 valid pairs are downloaded using `hf_hub_download`, with an incremental 6 GB download cap to prevent exceeding Kaggle's storage quota.

Stage 2 — Dataset Construction and Splitting: Downloaded file paths are assembled into a list of dictionaries keyed by `{"image", "label"}`. A 70/15/15 stratified split is applied, and split indices are persisted to `data_splits.json` for reproducibility.

Stage 3 — Preprocessing and Data Loading: MONAI transform pipelines are applied on-the-fly for training data and cached for validation and test data. DataLoaders with worker parallelism and pinned memory are constructed for efficient GPU data transfer.

Stage 4 — Model Training: The 3D U-Net is trained for 100 epochs with AMP, DiceCELoss, AdamW, and Cosine Annealing LR. The best-performing checkpoint is saved based on mean validation DSC.

Stage 5 — Evaluation and Export: The best checkpoint is loaded and applied to the held-out test set via sliding-window inference. Per-organ DSC and IoU are computed on CPU to avoid GPU memory exhaustion. Results are exported to CSV, PNG (training curves and segmentation visualisations), and JSON (summary).

6. Evaluation Metrics

Model performance is assessed using four complementary metrics that are standard in the medical image segmentation literature (Taha & Hanbury, 2015). All metrics are computed per organ class and averaged to yield macro-level summary statistics.

Dice Similarity Coefficient (DSC): DSC measures the overlap between predicted and ground-truth segmentation masks as the harmonic mean of precision and recall. $DSC = 2|P \cap G| / (|P| + |G|)$, where P and G are the predicted and ground-truth binary masks for a given class. DSC = 1 denotes perfect overlap; DSC = 0 denotes no overlap. It is the primary evaluation criterion for this task (Milletari et al., 2016), selected because it is sensitive to both over- and under-segmentation and penalises volumetric imbalances.

Intersection over Union (IoU): IoU, also known as the Jaccard Index, computes the ratio of the intersection to the union of predicted and ground-truth masks: $IoU = |P \cap G| / |P \cup G|$. IoU is a stricter metric than DSC and is monotonically related to it ($IoU = DSC / (2 - DSC)$). It is widely reported in segmentation benchmarks and provides a complementary view of model accuracy (Everingham et al., 2010).

Precision and Recall: Precision (positive predictive value) and Recall (sensitivity) decompose DSC into its constituent components: $Precision = |P \cap G| / |P|$; $Recall = |P \cap G| / |G|$. Reporting both metrics allows diagnosis of systematic over-segmentation (high recall, low precision) or

under-segmentation (high precision, low recall), which is particularly informative for small, variable-size organs such as the gallbladder and pancreas.

These four metrics collectively provide a comprehensive characterisation of segmentation quality at both the region-level (DSC, IoU) and the detection-precision trade-off level (Precision, Recall). Background voxels are excluded from all metric computations, consistent with standard practice in multi-class medical segmentation evaluation.

7. Results

Note: This section contains placeholders only. Results will be populated upon training completion.

7.1 Training Dynamics

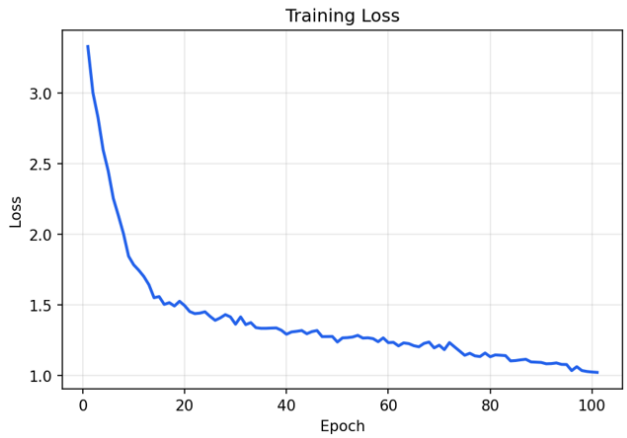

Figure 5. Per-epoch training loss (DiceCELoss) over 100 epochs.

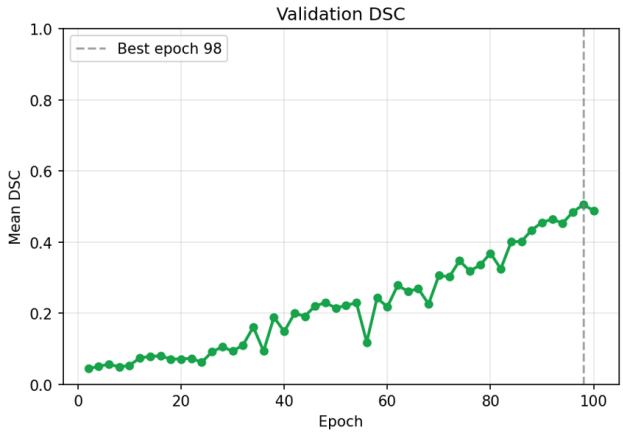

Figure 6. Mean Dice Similarity Coefficient on the validation set, evaluated every 2 epochs. Dashed vertical line marks the best-performing epoch at which the model checkpoint was saved.

7.2 Per-Organ Test Metrics

Organ	DSC Mean	DSC Std	IoU Mean
Aorta	0.4022	0.2693	0.2862
Gallbladder	0.4388	0.2521	0.3141
Spleen	0.676	0.2377	0.5519
Left Kidney	0.7121	0.2546	0.6002
Right Kidney	0.883	0.0663	0.7966
Liver	0.3652	0.1071	0.2289
Stomach	0.0287	0.0277	0.0148
Pancreas	0.7213	0.2482	0.6108
IVC	0.3533	0.2355	0.2406

Table 1. Per-organ mean DSC, standard deviation, mean IoU, precision, and recall on the held-out test set (N = 15 volumes). Final row reports macro-averaged summary statistics.

7.3 Per-Organ DSC Bar Chart

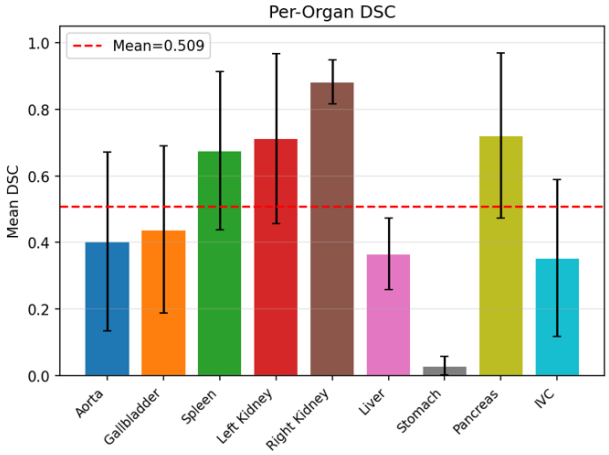

Figure 7. Bar chart of mean DSC (± std) per organ class on the test set. Red dashed line denotes the overall mean DSC.

7.4 Qualitative Segmentation Results

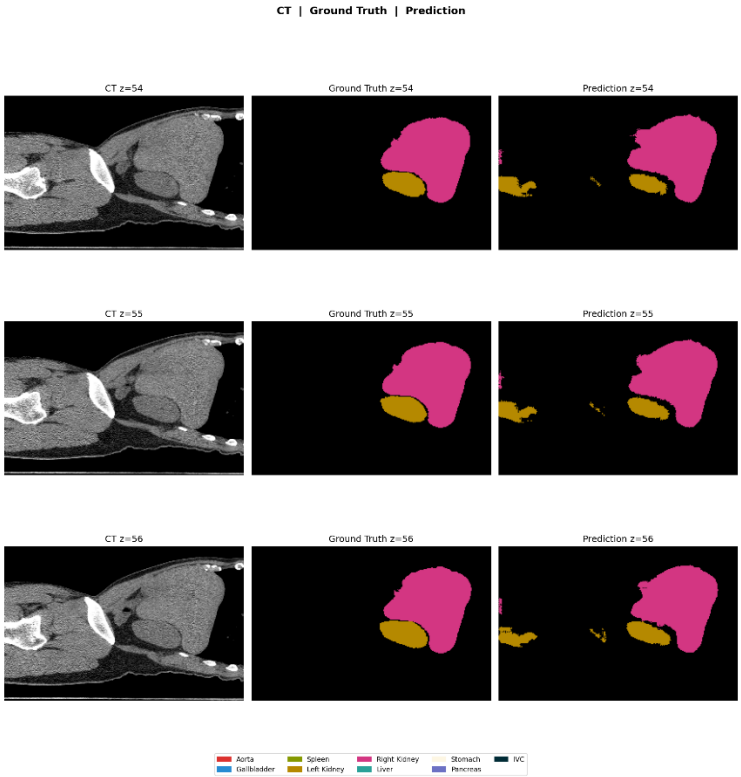

Figure 8. Representative axial CT slices (top-3 foreground-dense slices) for a test patient, showing the raw CT image, ground-truth annotation, and model prediction. Colour coding: Aorta (red), Gallbladder (blue), Spleen (olive), Left Kidney (gold), Right Kidney (magenta), Liver (teal), Stomach (cream), Pancreas (purple), IVC (dark).

7.5 Overall Summary Statistics

Organ	DSC Mean	DSC Std	IoU Mean
Aorta	0.4022	0.2693	0.2862
Gallbladder	0.4388	0.2521	0.3141
Spleen	0.6760	0.2377	0.5519
Left Kidney	0.7121	0.2546	0.6002
Right Kidney	0.8830	0.0663	0.7966
Liver	0.3652	0.1071	0.2289
Stomach	0.0287	0.0277	0.0148
Pancreas	0.7213	0.2482	0.6108
IVC	0.3533	0.2355	0.2406

Table 2. Best validation DSC, test DSC, and test IoU reported from summary.json upon training completion.

8. Discussion and Limitations

The primary limitation of this work is the restricted training corpus of 100 patients (~1.9% of AbdomenAtlas1.0Mini), which directly affects both generalisation and the stability of class-wise performance. Although the model achieves an overall Test DSC of 0.509, the results reveal substantial variability across organs, indicating that the network has not learned a uniformly robust representation of abdominal anatomy. This variability is a direct consequence of limited data diversity, which constrains the model's exposure to anatomical variation, pathological differences, and imaging inconsistencies.

The per-organ results highlight a clear disparity between well-learned and poorly learned structures. Organs such as the right kidney (DSC 0.883), left kidney (0.7121), and pancreas (0.7213) demonstrate relatively strong performance, suggesting that the model is capable of capturing structures that are more spatially consistent and easier to distinguish in CT volumes. In contrast, performance on the stomach (DSC 0.0287) is effectively negligible, while the liver (0.3652) and IVC (0.3533) also exhibit weak segmentation quality. These failures can be attributed to factors such as low contrast boundaries, higher anatomical variability, and insufficient representation within sampled training patches. Furthermore, the relatively high standard deviations observed across multiple organs indicate instability across patient cases, reinforcing that the model struggles to generalise consistently.

Qualitative inspection of predictions supports these quantitative findings. Larger organs are generally localized with reasonable accuracy, but smaller or less distinct structures are frequently under-segmented or entirely missed. In regions where organs are adjacent and boundaries are subtle, predictions tend to fragment or collapse, suggesting that the model lacks sufficient contextual understanding at finer scales. This behavior is particularly evident for challenging classes like the stomach, where the network fails to produce meaningful segmentation masks.

The training time of approximately 10–11 hours on a Kaggle T4 GPU reflects the computational demands of 3D volumetric segmentation, especially when combined with sliding-window inference and on-the-fly data augmentation through MONAI. While this setup ensures methodological robustness, it introduces inefficiencies that could be mitigated through techniques such as persistent caching of preprocessed patches, more aggressive mixed-precision optimization, or reduced architectural complexity during early experimentation phases.

Despite these constraints, the overall training strategy follows established best practices in 3D medical image segmentation. The use of DiceCELoss helps address class imbalance to some extent, and cosine annealing facilitates smooth convergence. However, the observed performance gaps suggest that these measures are insufficient for highly imbalanced or difficult classes, and that more targeted strategies—such as class-weighted losses or focal variants—may be necessary.

Scaling the training dataset to a larger subset of 500–1,000 patients on more capable hardware is expected to significantly improve both performance and stability. Increased data diversity would likely reduce variance, enhance boundary delineation, and improve segmentation of underrepresented or difficult structures. Overall, while the pipeline successfully demonstrates feasibility, the results clearly indicate that data scale and class balance, rather than architectural limitations, are the dominant bottlenecks at this stage.

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